

CAT Installation Guide

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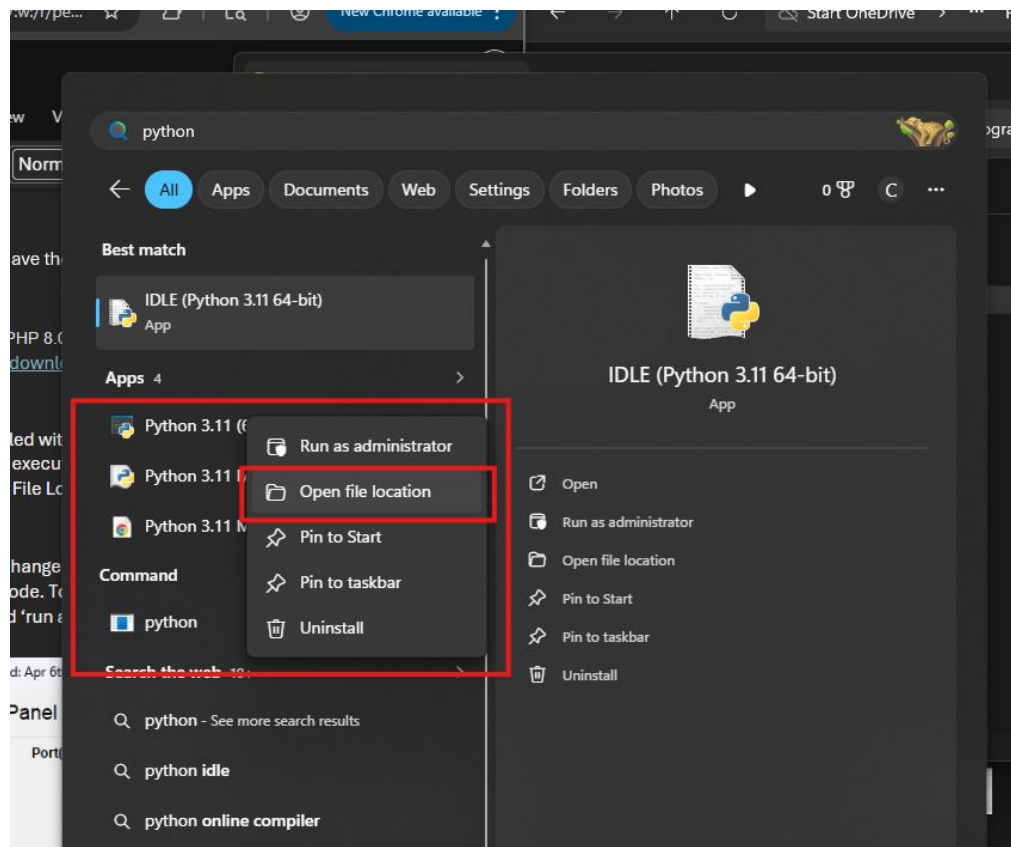
Pre-requisites

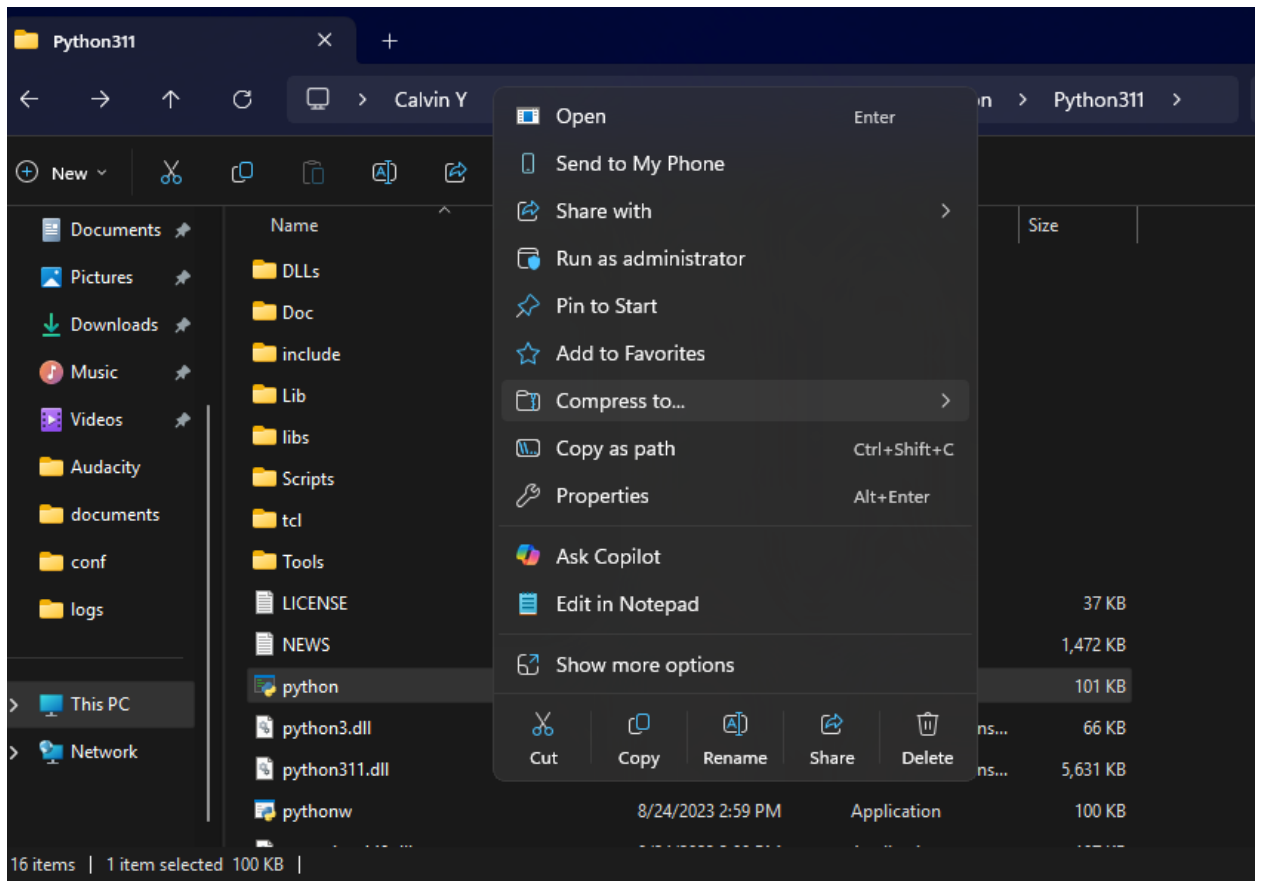
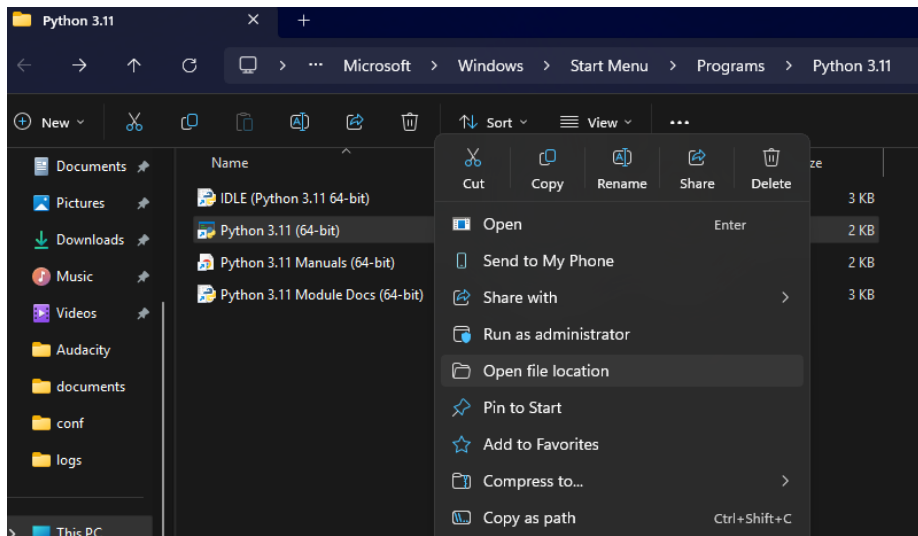
Before proceeding, please ensure you have the “htdocs” source code downloaded, the repository open, and the following Applications downloaded:

1. (XAMPP for Windows: 8.0.30 / PHP 8.0.30) Download XAMPP from:

<https://www.apachefriends.org/download.html>

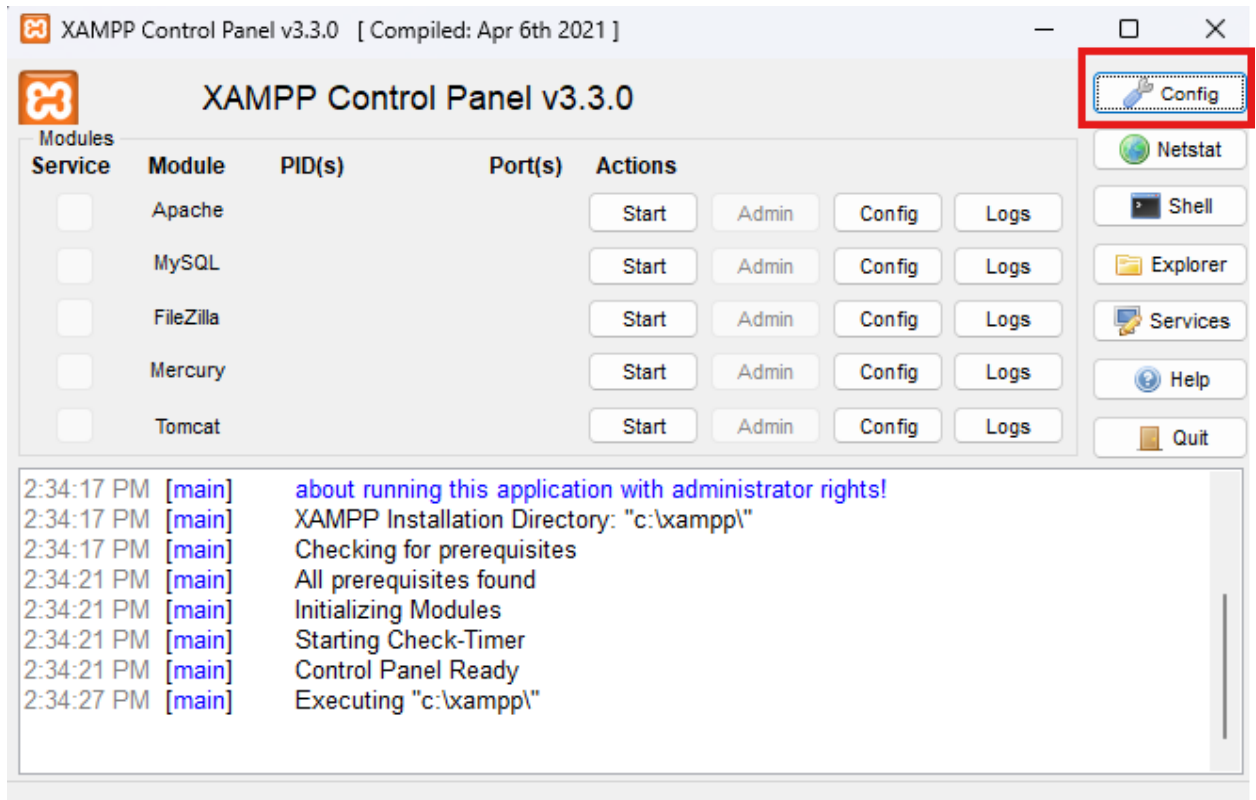
2. Ensure you have Python installed with versions less than 3.13 and above or equal to 3.10. Please note your Python executable path. This can be found by right clicking your Python application -> Open File Location -> right click Python 3.xx (64-bit) -> Open File location -> right click python.exe and copy as path:



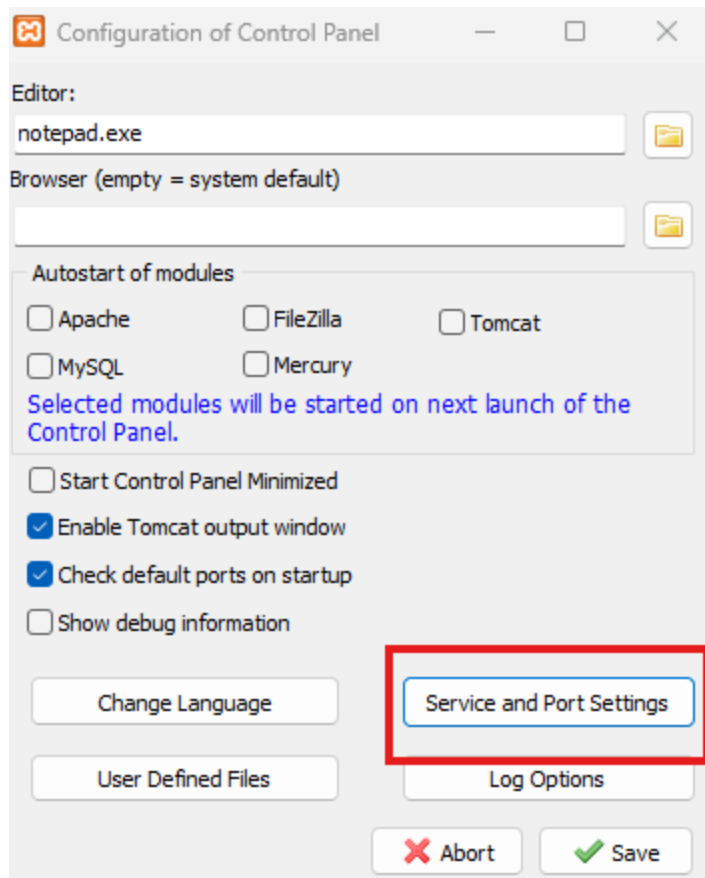


XAMPP Initial Configuration

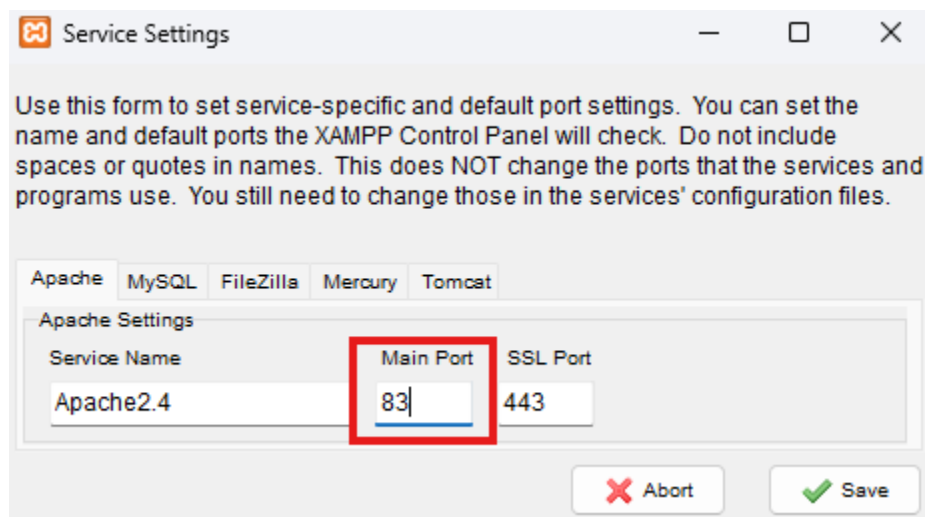
1. Initial setup may require you to change XAMPP default ports. Run the XAMPP control panel in Administrator Mode. To do this, before opening the 'xampp control panel' application, right click and 'run as administrator'. Then, select the config box as shown below.



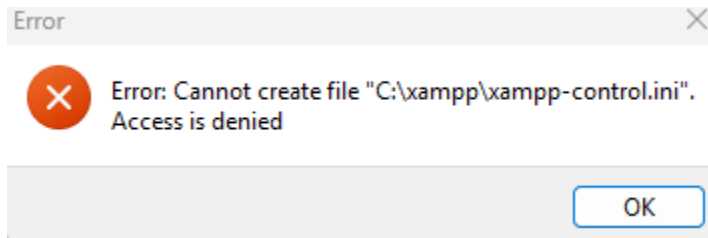
2. Go to 'Service and Port Settings'.



3. Change the Main Port for Apache to something other than 80. Do the same for the MySQL port by clicking the MySQL tab next to the Apache tab in the left corner.

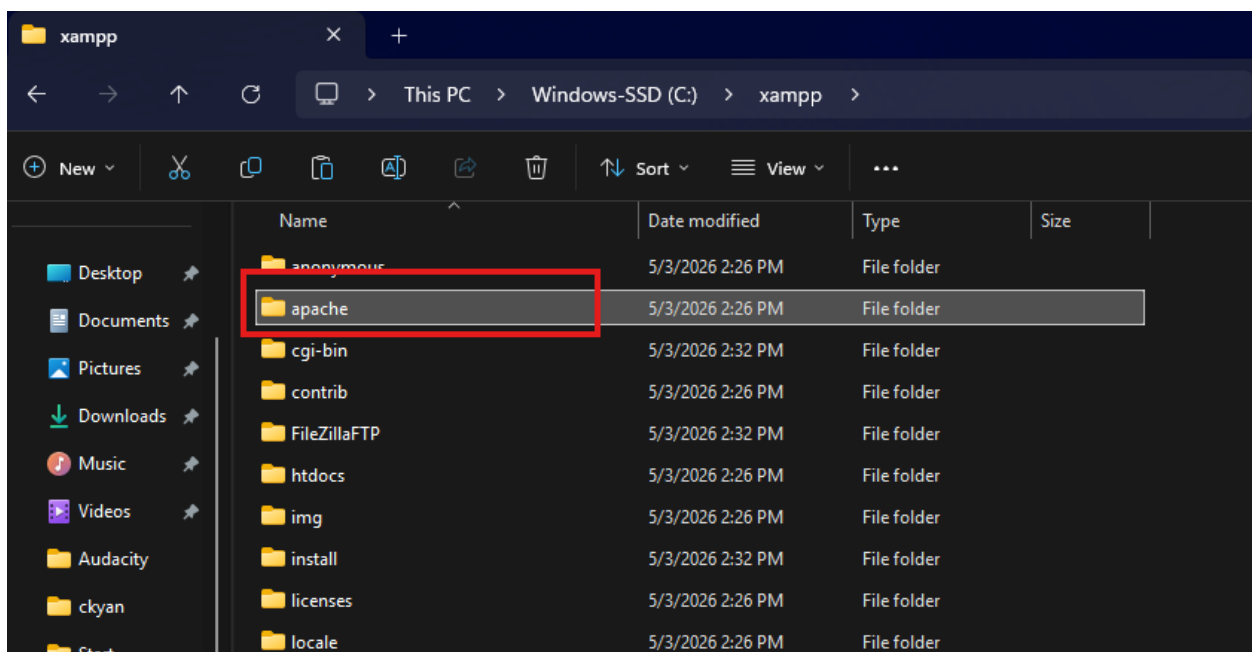


3. If you get this error message, you may not be in administrator mode. Restart your xampp control panel in Administrator Mode.

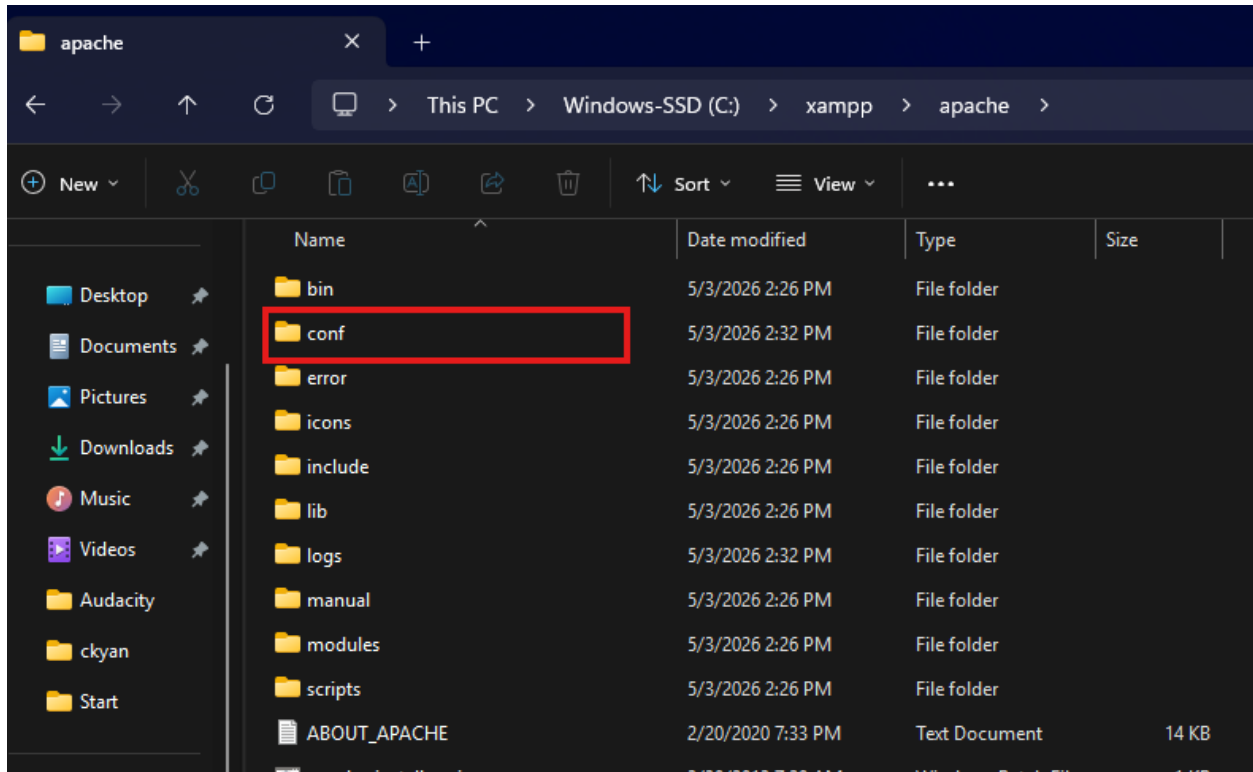


File and File Path Changes

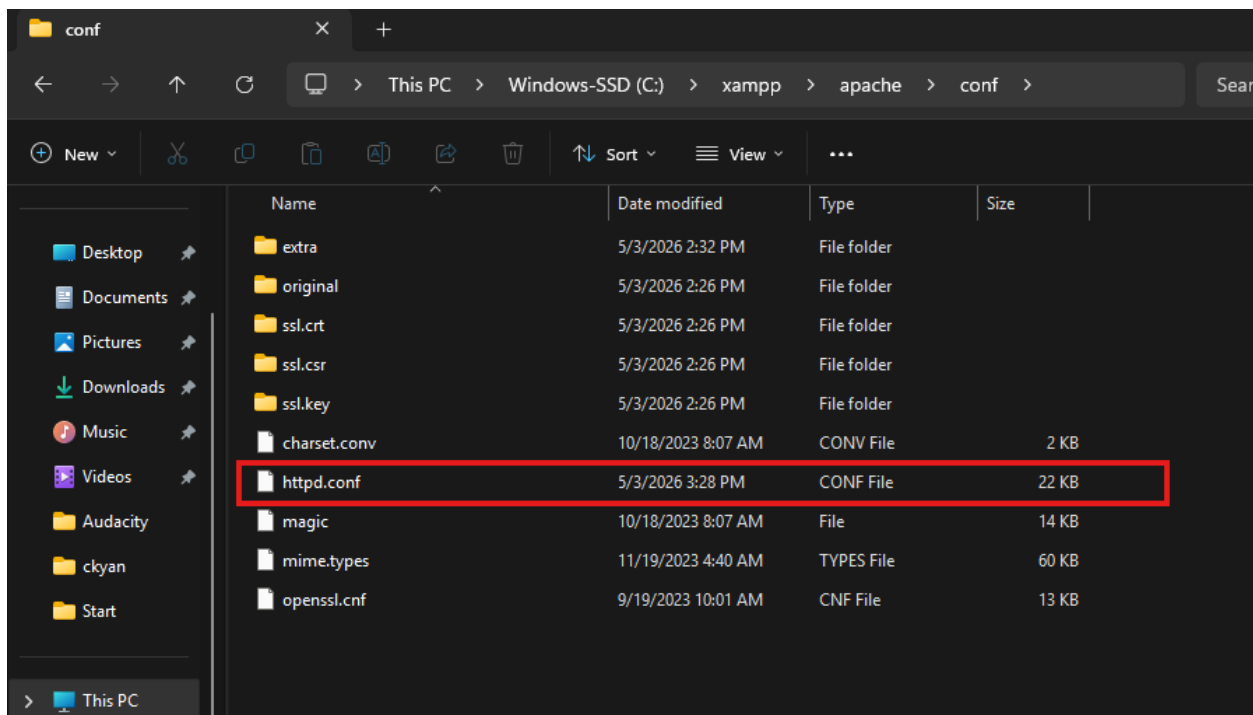
4. We will now make a series of edits to the files and file paths in the source code. First, ensure that you have the downloaded and extracted “htdocs” file attached to the submission. Then, locate the apache folder within your xampp folder.



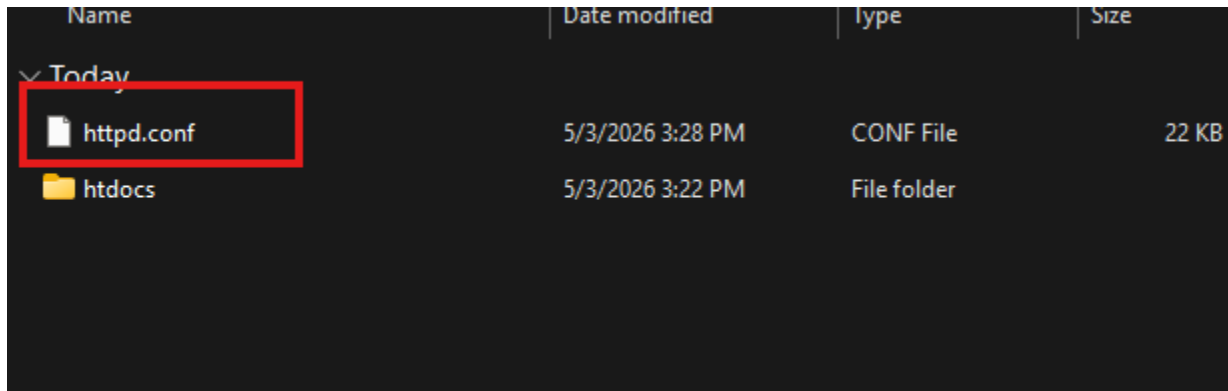
5. Then head into the “conf” folder



6. Note the “httpd.conf” file for the following step.

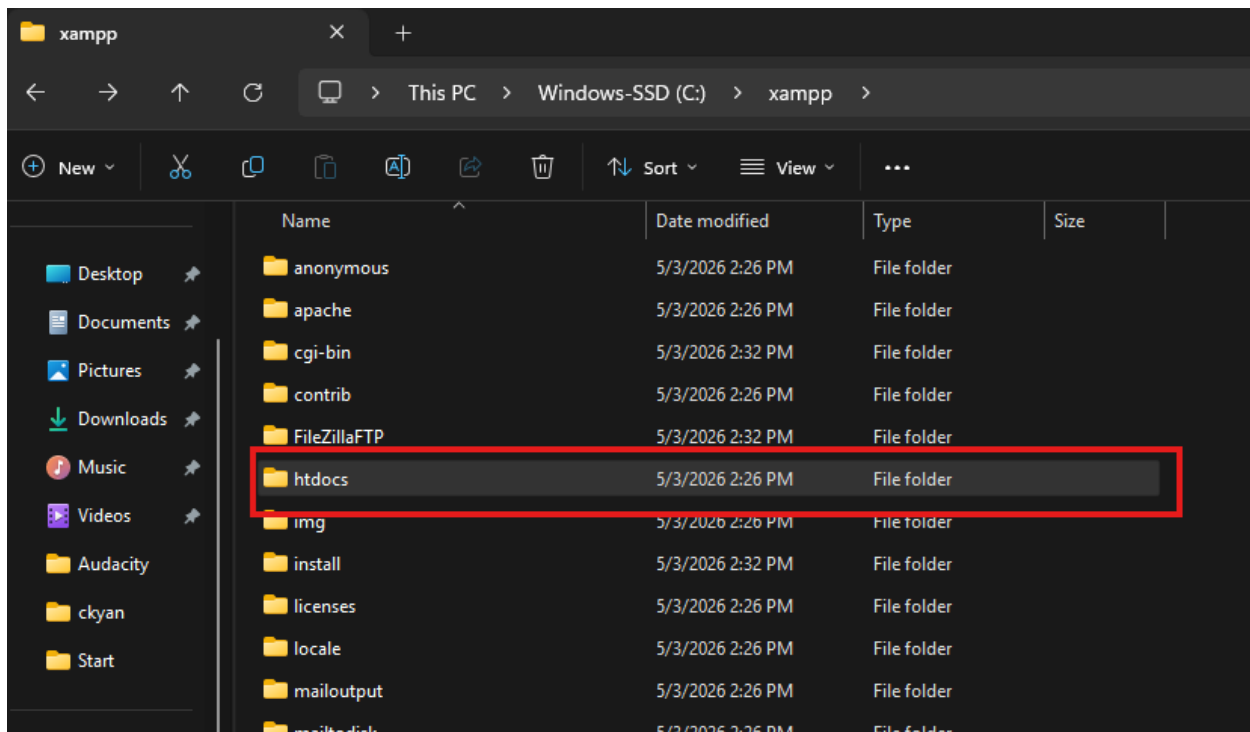


7. In the extracted htdocs folder, find the “httpd.conf” file. Drag and drop this file and replace the existing “httpd.conf” file from the previous step.



Name	Date modified	Type	Size
Today			
httpd.conf	5/3/2026 3:28 PM	CONF File	22 KB
htdocs	5/3/2026 3:22 PM	File folder	

8. Go back to the root directory of your xampp folder. Find and delete the “htdocs” folder. Then, as noted in the previous step, drag and drop the extracted “htdocs” folder into the directory.

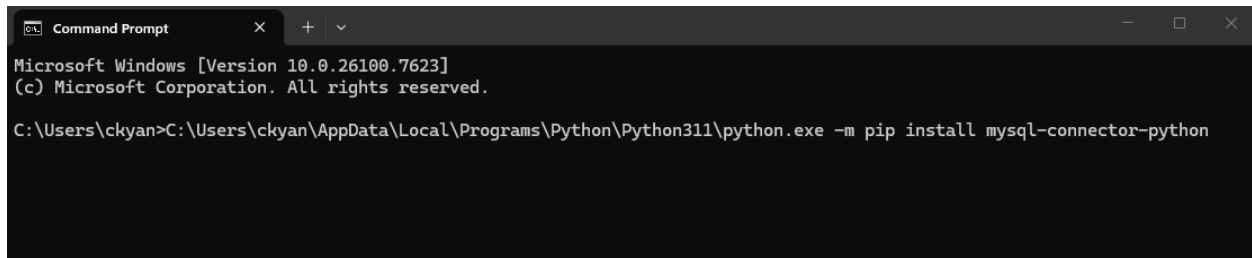


Name	Date modified	Type	Size
anonymous	5/3/2026 2:26 PM	File folder	
apache	5/3/2026 2:26 PM	File folder	
cgi-bin	5/3/2026 2:32 PM	File folder	
contrib	5/3/2026 2:26 PM	File folder	
FileZillaFTP	5/3/2026 2:32 PM	File folder	
htdocs	5/3/2026 2:26 PM	File folder	
img	5/3/2026 2:26 PM	File folder	
install	5/3/2026 2:32 PM	File folder	
licenses	5/3/2026 2:26 PM	File folder	
locale	5/3/2026 2:26 PM	File folder	
mailoutput	5/3/2026 2:26 PM	File folder	
mailtodisk	5/3/2026 2:26 PM	File folder	

Install Python SQL Connector

9. To ensure that SQL is installed globally, find your python executable path, replace the path listed and run the following command in your command prompt:

Your_python_executable_path\python.exe -m pip install mysql-connector-python

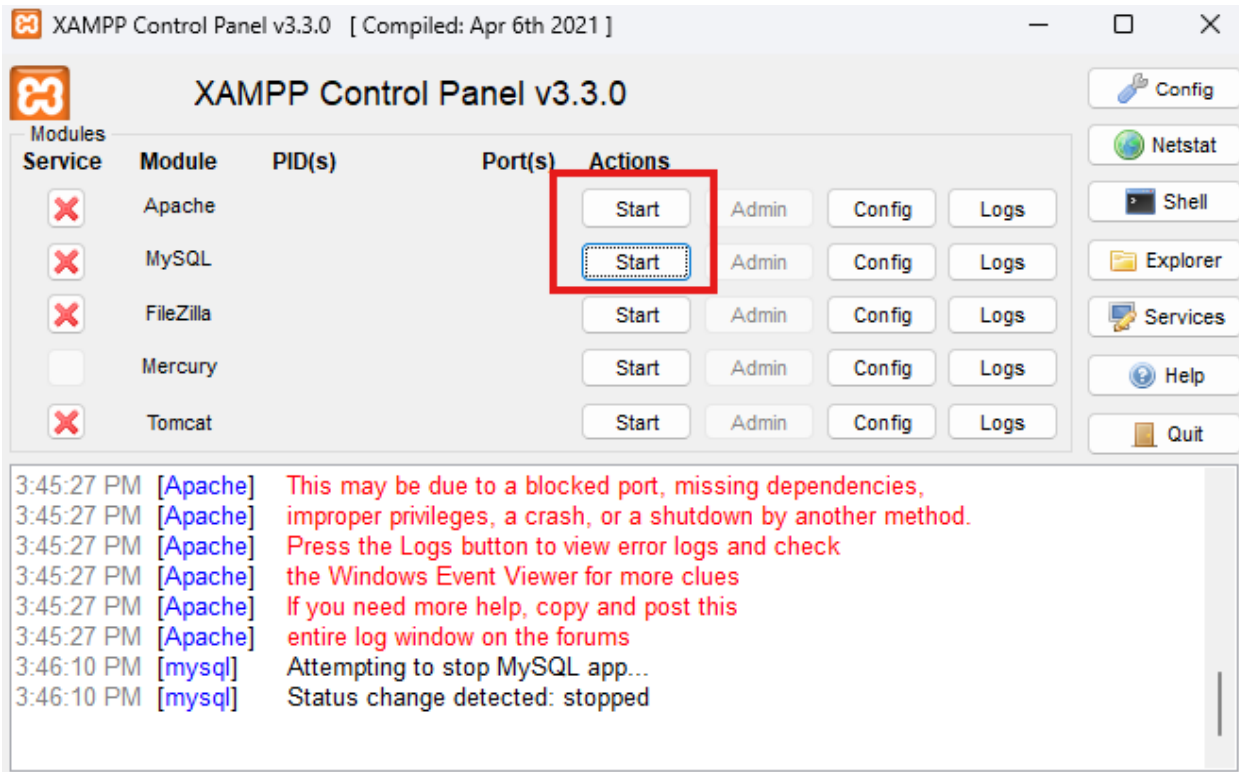


```
Command Prompt
Microsoft Windows [Version 10.0.26100.7623]
(c) Microsoft Corporation. All rights reserved.

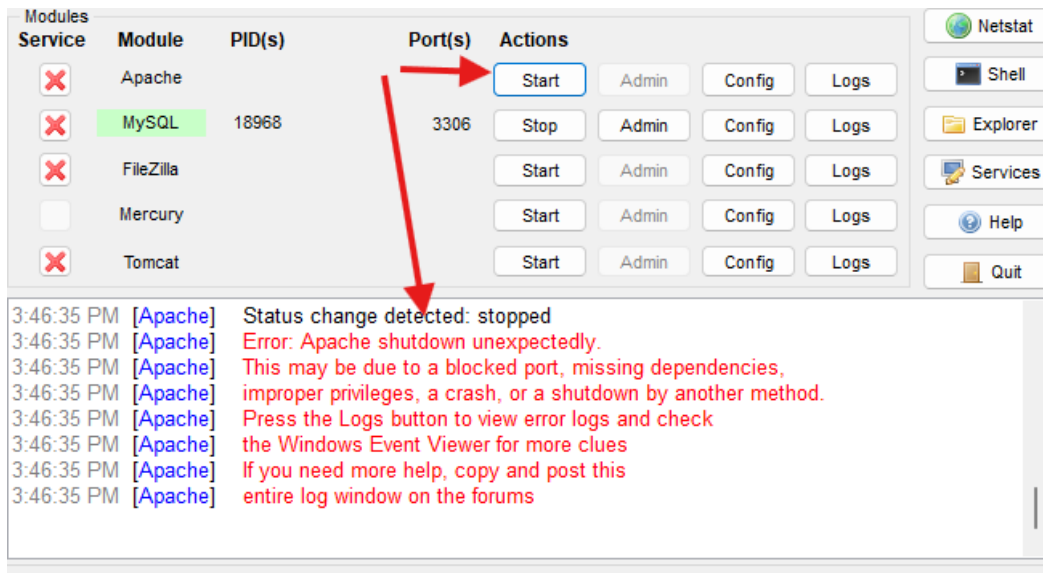
C:\Users\ckyan>C:\Users\ckyan\AppData\Local\Programs\Python\Python311\python.exe -m pip install mysql-connector-python
```

Initialization Errors

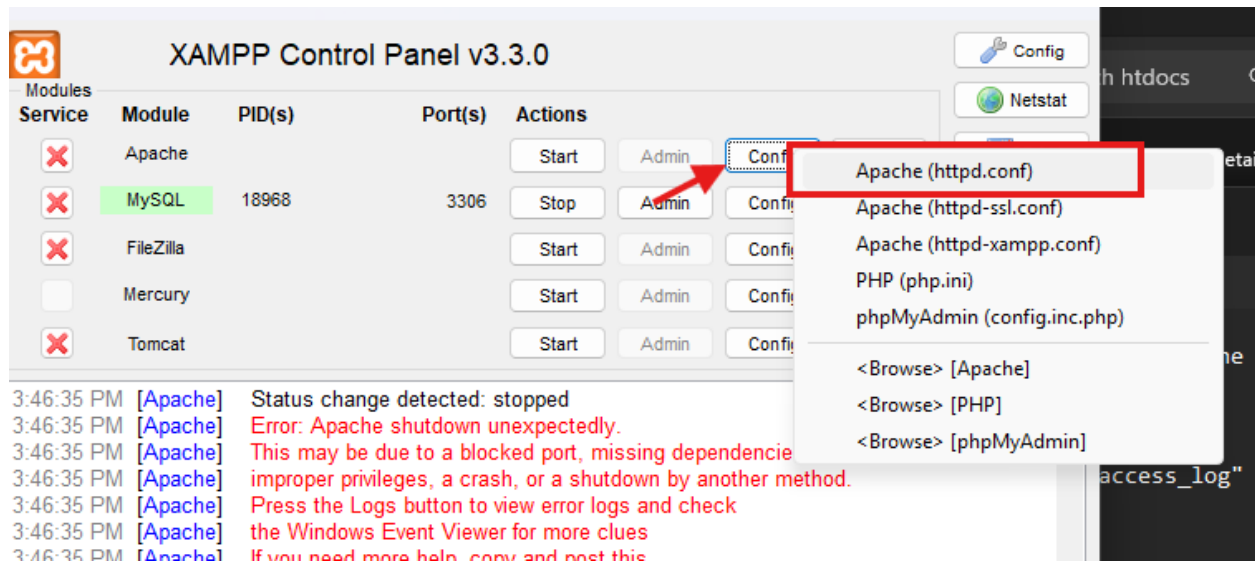
10. Now, start both servers in the XAMPP control panel.



10. 2. You may encounter the following errors:



10.3. In the control panel, click on “Config” next to the Apache row. Head into the httpd.conf file.



10.4. In your “httpd.conf” file, replace the following “D:/ics499” paths that appear throughout the file with your xampp apache location.

a.

```

File Edit View

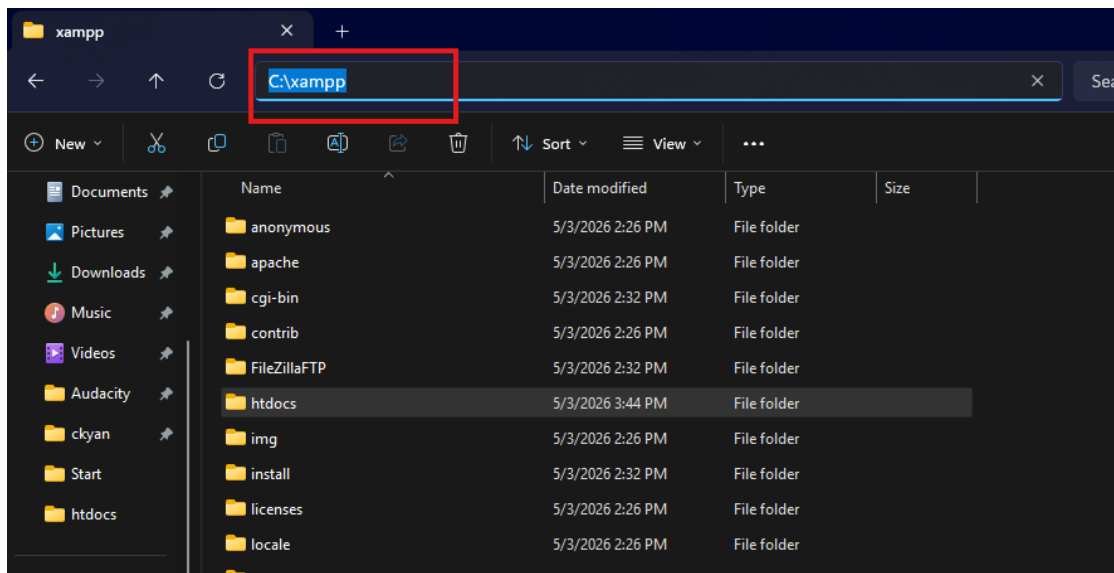
# will be used by default. It is recommended that you always supply
# an explicit drive letter in absolute paths to avoid confusion.

#
# ServerRoot: The top of the directory tree under which the server's
# configuration, error, and log files are kept.
#
# Do not add a slash at the end of the directory path. If you point
# ServerRoot at a non-local disk, be sure to specify a local disk on the
# Mutex directive, if file-based mutexes are used. If you wish to share the
# same ServerRoot for multiple httpd daemons, you will need to change at
# least PidFile.
#
Define SRVROOT "D:/ics499/apache"
ServerRoot "D:/ics499/apache"

#
# Mutex: Allows you to set the mutex mechanism and mutex file directory
# for individual mutexes, or change the global defaults
#
# Uncomment and change the directory if mutexes are file-based and the default
# mutex file directory is not on a local disk or is not appropriate for some
# other reason.
#
# Mutex default:logs

```

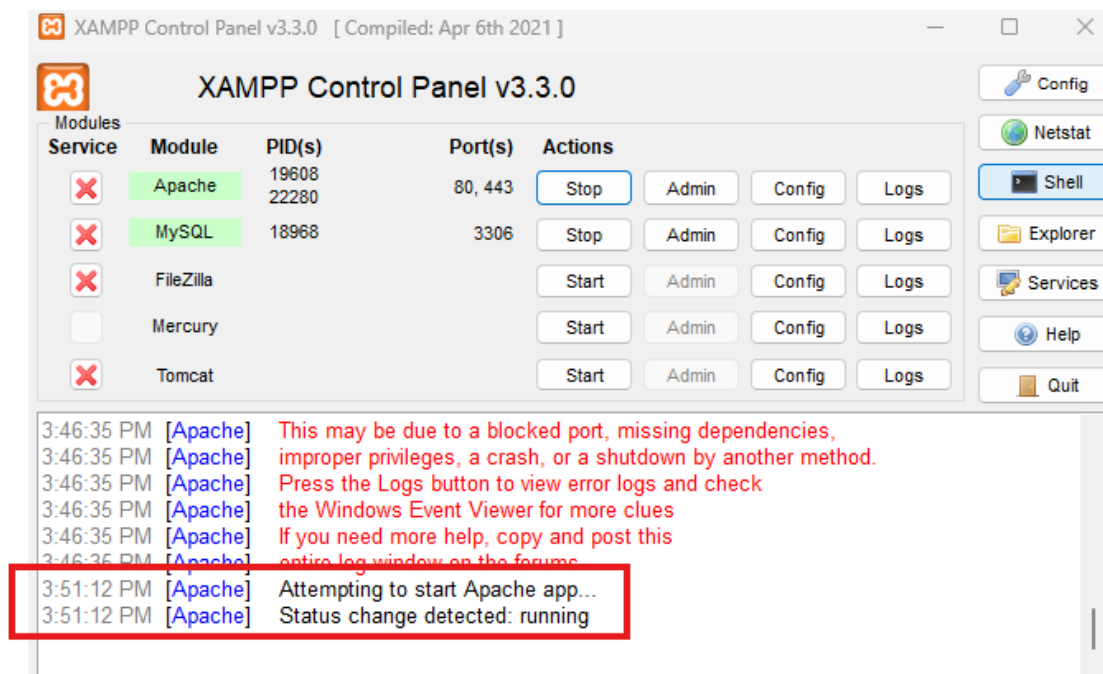
b.



c.

```
error.log httpd.conf
File Edit View
# will be used by default. It is recommended that you always supply
# an explicit drive letter in absolute paths to avoid confusion.
#
# ServerRoot: The top of the directory tree under which the server's
# configuration, error, and log files are kept.
#
# Do not add a slash at the end of the directory path. If you point
# ServerRoot at a non-local disk, be sure to specify a local disk on the
# Mutex directive, if file-based mutexes are used. If you wish to share the
# same ServerRoot for multiple httpd daemons, you will need to change at
# least PidFile.
#
Define SRVROOT "C:/xampp/apache"
ServerRoot "C:/xampp/apache"
#
# Mutex: Allows you to set the mutex mechanism and mutex file directory
# for individual mutexes, or change the global defaults
#
# Uncomment and change the directory if mutexes are file-based and the default
# mutex file directory is not on a local disk or is not appropriate for some
# other reason.
#
# Mutex default:log
```

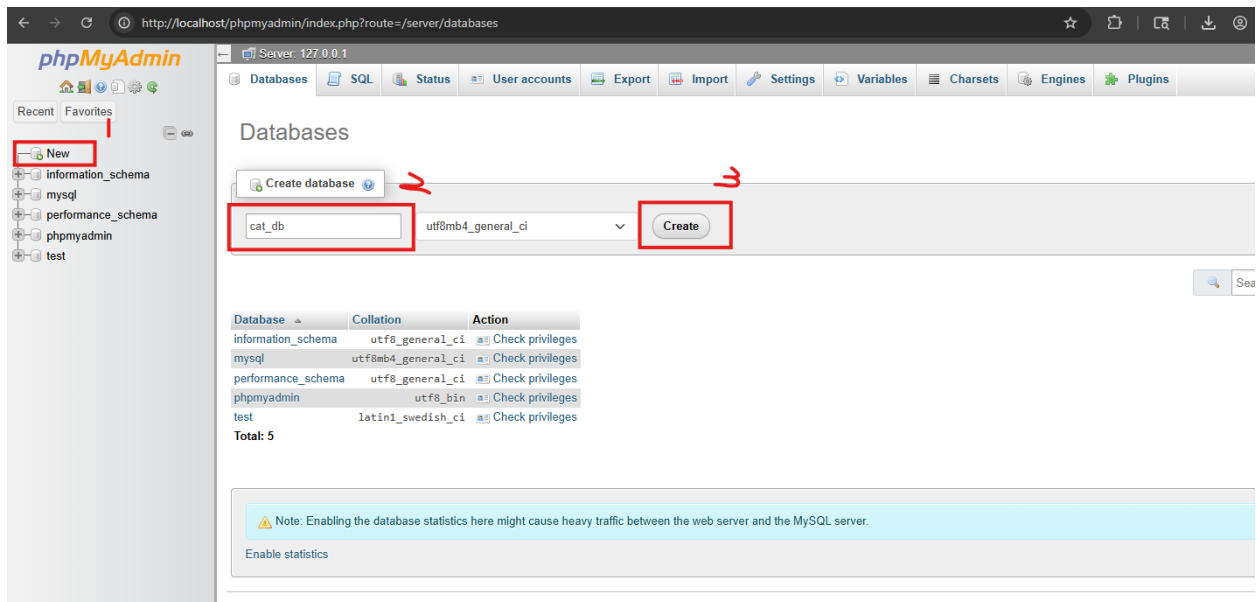
10.4. Apache should now run.



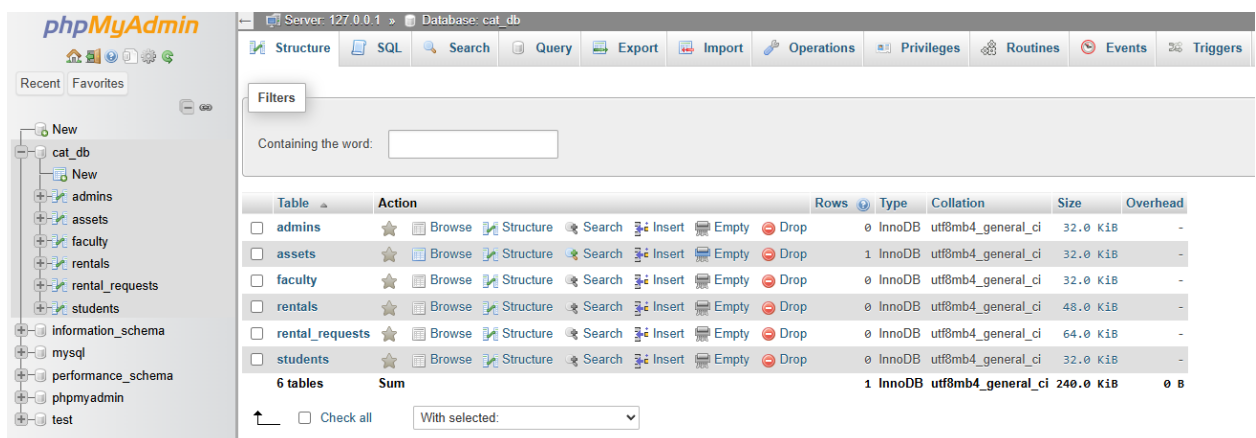
Database Setup and Initialization – phpMyAdmin

All backend database management will be done here. Any direct and hard edits may be made using this page.

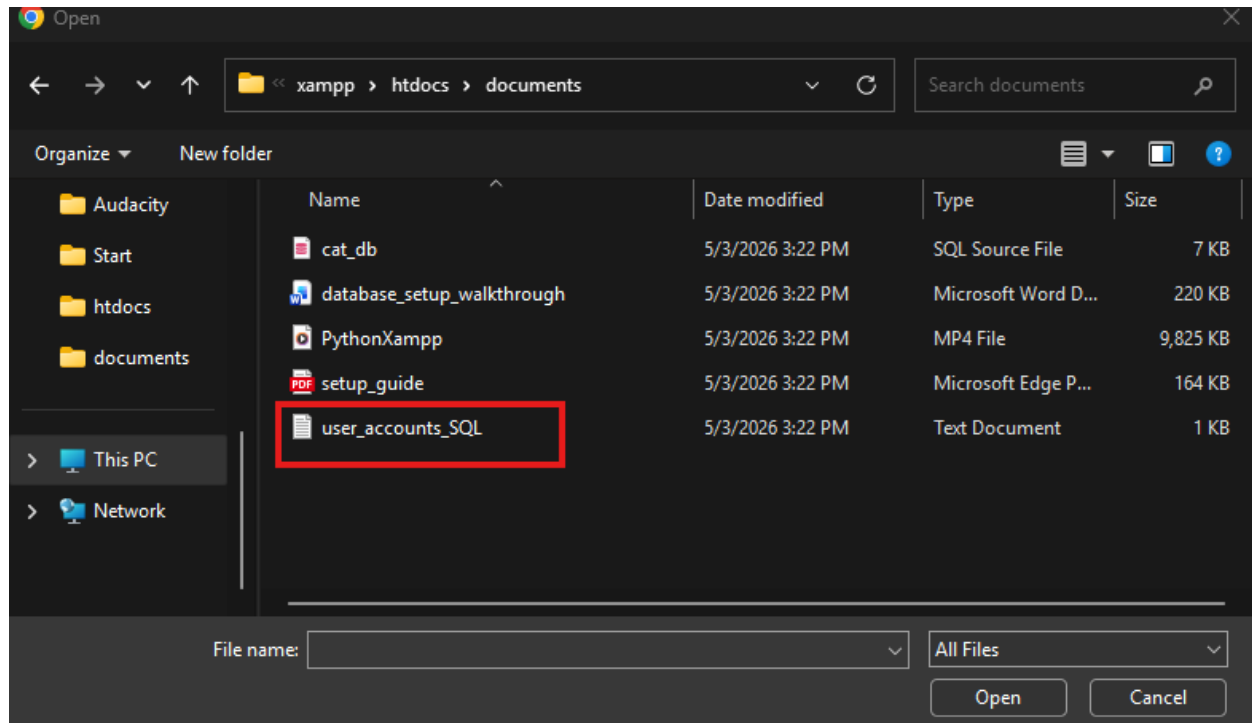
11. Head to: <http://localhost/phpmyadmin/index.php>. On the left side, click on “New”. In the name field, type in “cat_db”. Then, click create.



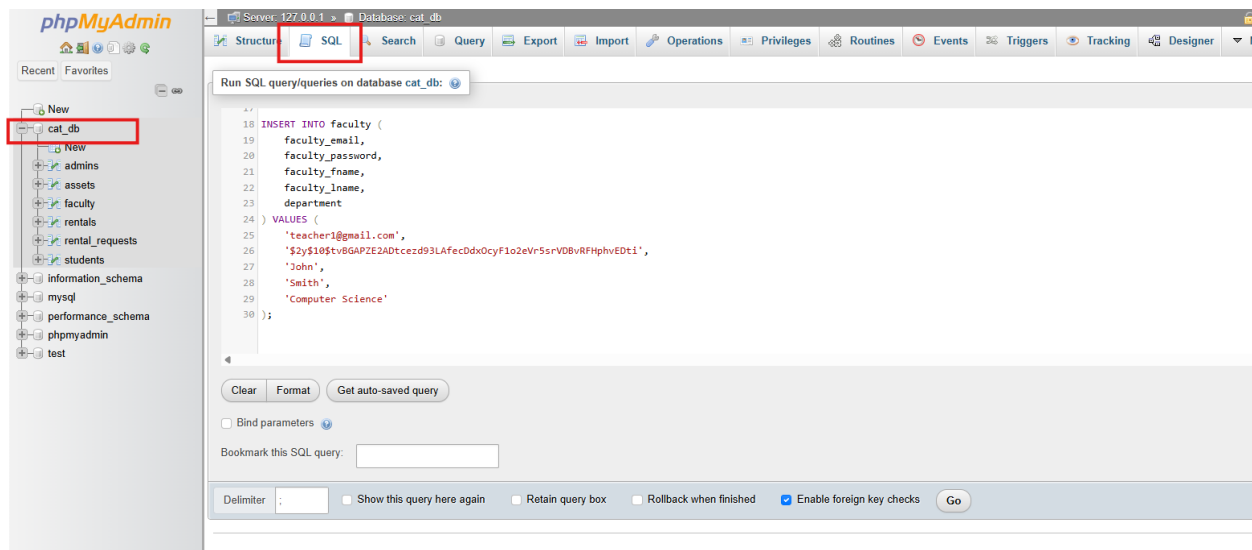
12. Now, follow import directions listed in the repository’s document folder titled: ‘database_setup_walkthrough.docx’. The “cat_db.sql” file will be found in the same “documents” location in the repository. Once the database is imported, you should see the following:



13. We will now import the base users. In the same location, you see will a “user_accounts_SQL” file.



14. Now, in the phpMyAdmin page, click on the cat_db database on the left. Then, click on the SQL tab as shown below. Copy and paste the contents of the file into the sql tab. Then, press go on the bottom right. The accounts should now be registered.:



Runtime Errors

15. You will now run into these errors.

Internal Server Error

The server encountered an internal error or misconfiguration and was unable to complete your request.

Please contact the server administrator at postmaster@localhost to inform them of the time this error occurred, and the actions you performed just before this error.

More information about this error may be available in the server error log.

Apache/2.4.58 (Win64) OpenSSL/3.1.3 PHP/8.0.30 Server at localhost Port 80

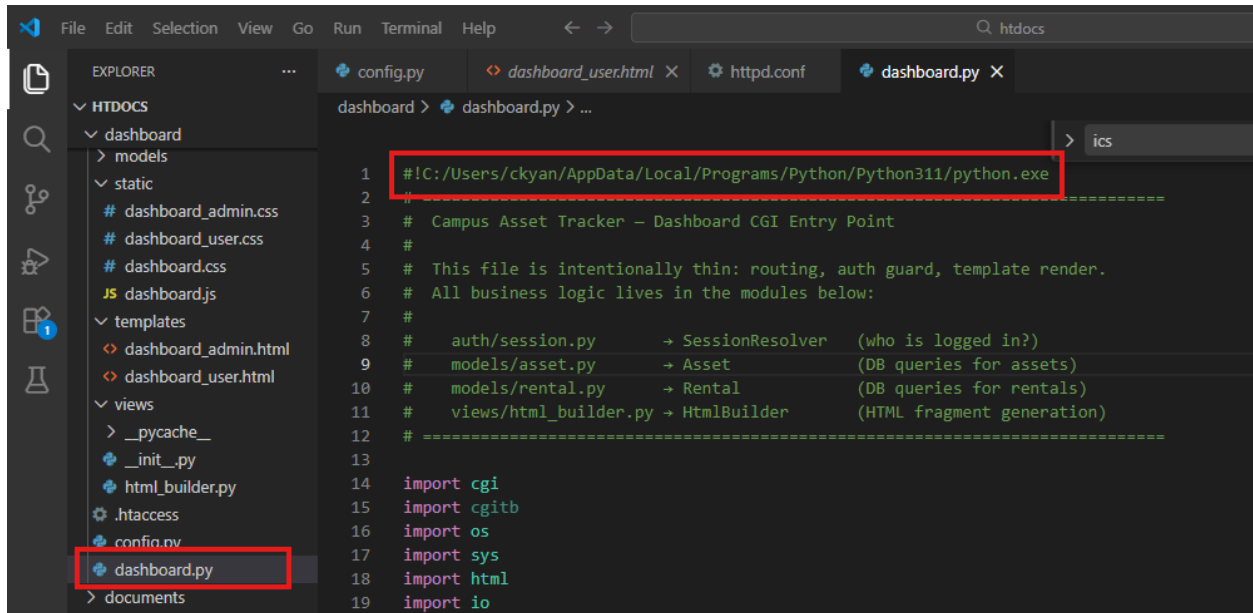
16. Open the following files and edit the paths. The first file we will edit can be found in your xampp's htdocs folder. Head to: *your_xampp_path\htdocs\dashboard*. Find the config.py file and change the following variable. Change the listed path to *"your_xampp_path/tmp"*


```
File Edit Selection View Go Run Terminal Help
config.py
74 DEFAULT_STATUSES = [
75     "All Statuses",
76     "available",
77     "in-use",
78     "maintenance",
79 ]
80
81 # Rental request filter options shown to admin
82 RENTAL_REQUEST_FILTERS = ["Pending", "Approved", "Denied", "All"]
83
84 # -----
85 # Session
86 # -----
87 SESSION_COOKIE_NAME = "cat session token"
88 PHP_SESSION_PATH = "D:/ics499/tmp" # Update this to match your xampp's tmp file path
89 PHPSESSID_COOKIE = "PHPSESSID"
90
91 # -----
92 # Roles
93 # -----
94 ROLES = ["admin", "faculty", "student"]
95 ADMIN_ROLES = ["admin"]
96 USER_ROLES = ["faculty", "student"]
```

17. Then, find the “dashboard.py” file. Find the following line and delete or comment it out.

```
File Edit Selection View Go Run Terminal Help
dashboard.py
10 # models/rental.py -> Rental (DB queries for rent)
11 # views/html_builder.py -> HtmlBuilder (HTML fragment gener)
12 # -----
13
14 import cgi
15 import cgitb
16 import os
17 import sys
18 import html
19 import io
20 import json
21
22 sys.stdout = io.TextIOWrapper(sys.stdout.buffer, encoding="utf-8")
23 cgitb.enable(display=0, logdir="C:/xampp/tmp")
24
25 import config
26 from auth.session import SessionResolver
27 from models.asset import Asset
28 from models.rental import Rental
29 from views.html_builder import HtmlBuilder
30
31 import warnings
32 warnings.filterwarnings("ignore")
33
34 # HTTP UTILITIES
35 def redirect(url: str):
```

18. Next, find the first line at the top of the “dashboard.py” file. With your noted python file path, replace this entire line. Ensure that there are no quotation marks and the start or end of your path.

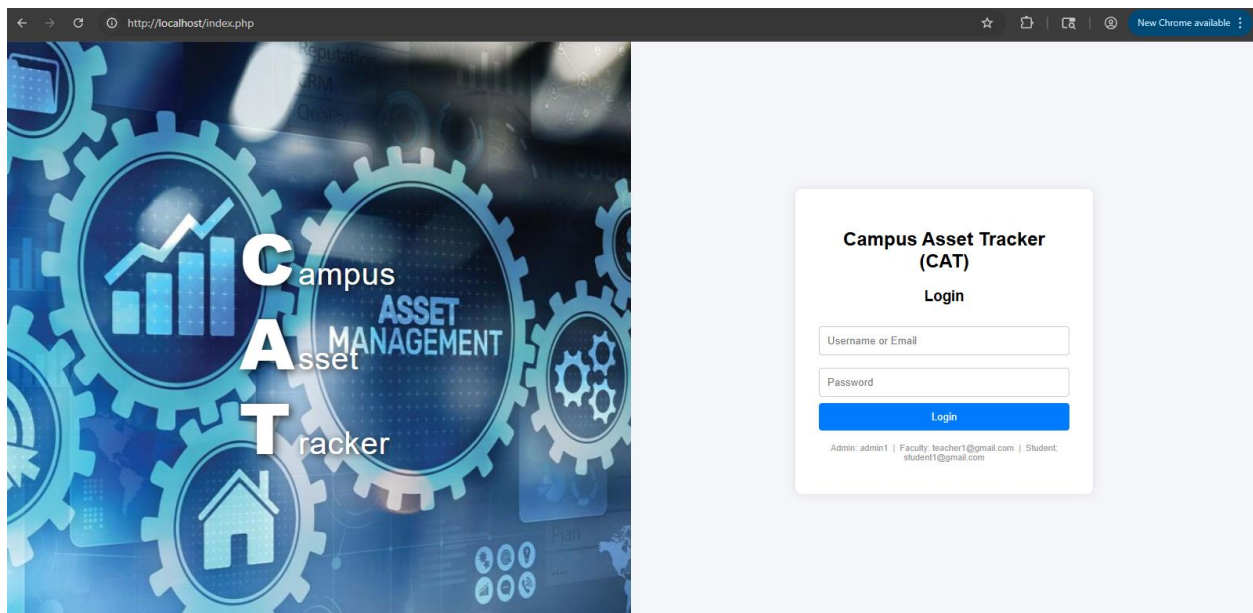


```
1 #!C:/Users/ckyan/AppData/Local/Programs/Python/Python311/python.exe
2 #
3 # Campus Asset Tracker – Dashboard CGI Entry Point
4 #
5 # This file is intentionally thin: routing, auth guard, template render.
6 # All business logic lives in the modules below:
7 #
8 #   auth/session.py      → SessionResolver   (who is logged in?)
9 #   models/asset.py      → Asset              (DB queries for assets)
10 #   models/rental.py     → Rental             (DB queries for rentals)
11 #   views/html_builder.py → HtmlBuilder       (HTML fragment generation)
12 # =====
13
14 import cgi
15 import cgitb
16 import os
17 import sys
18 import html
19 import io
```

19. Now, you should be able to go to your browser and head to: <http://localhost/index.php>

Test accounts are listed below with the following format of *username password*:

- admin1 admin
- student1@gmail.com student
- teacher1@gmail.com teacher



Using The Application

Admin Logon

Once the admin user logs on, they will be able to view the admin dashboard. The total assets and their status will be displayed with stat cards. Below will be a detailed table listing the items and their metadata information.

 **Campus Asset Tracker**
IT Management Portal

admin1AdminManage UsersLogout

Asset Inventory

Manage all campus assets and rental requests

[Inventory](#)[Rental Management](#)[Summary](#)

Total Assets
 1

Available
 1

In Use
 0

Maintenance
 0

All Assets

All CategoriesAll StatusesSort

ID	NAME	CATEGORY	SERIAL NUMBER	STATUS	LOCATION
1	 Laptop1	IT equipment	1234ABCDE	Available	Computer Lab

Showing 1 of 1 assets

In the table, admins will also be able to dynamically change the status of items as they see fit by clicking on the status and selecting a status from the dropdown menu.

The screenshot shows the 'Asset Inventory' section of the 'Campus Asset Tracker' dashboard. The dashboard includes a header with the logo, user information (admin1, Admin), and links for 'Manage Users' and 'Logout'. The main content area has a sidebar with 'Inventory', 'Rental Management', and 'Summary' tabs. The 'Inventory' tab is active, showing a summary of assets: Total Assets (1), Available (1), In Use (0), and Maintenance (0). Below this is a table titled 'All Assets' with columns: ID, NAME, CATEGORY, SERIAL NUMBER, STATUS, and LOCATION. The first row shows an asset with ID 1, NAME 'Laptop1', CATEGORY 'IT equipment', SERIAL NUMBER '1234ABCDE', STATUS 'Available', and LOCATION 'Computer Lab'. A dropdown menu is open for the 'STATUS' column of the first row, showing options: Available (selected), In Use, Maintenance, and Reserved. The 'Add New Asset' button is visible in the top right corner.

ID	NAME	CATEGORY	SERIAL NUMBER	STATUS	LOCATION
1	Laptop1	IT equipment	1234ABCDE	Available	Computer Lab

Admins can also insert a new asset via the “Add New Asset” button.

The screenshot shows the 'Asset Inventory' section of the 'Campus Asset Tracker' dashboard. The dashboard includes a header with the logo, user information (admin1, Admin), and links for 'Manage Users' and 'Logout'. The main content area has a sidebar with 'Inventory', 'Rental Management', and 'Summary' tabs. The 'Inventory' tab is active, showing a summary of assets: Total Assets (2), Available (2), In Use (0), and Maintenance (0). Below this is a table titled 'All Assets' with columns: ID, NAME, CATEGORY, SERIAL NUMBER, STATUS, and LOCATION. The first row shows an asset with ID 1, NAME 'Laptop1', CATEGORY 'IT equipment', SERIAL NUMBER '1234ABCDE', STATUS 'Available', and LOCATION 'Computer Lab'. The second row shows an asset with ID 2, NAME 'Iphone', CATEGORY 'Technology', SERIAL NUMBER, STATUS 'Available', and LOCATION 'Test'. The 'Add New Asset' button is highlighted with a red box in the top right corner.

ID	NAME	CATEGORY	SERIAL NUMBER	STATUS	LOCATION
1	Laptop1	IT equipment	1234ABCDE	Available	Computer Lab
2	Iphone	Technology		Available	Test

This will lead them to an asset creation page and prompt for the asset name, category, location, and status. Once created, a success message will appear. Once reloading the dashboard, the asset will appear in the table.

Campus Asset Tracker

IT Management Portal

admin1

Admin

Manage Users

Logout

Inventory

Rental Management 1

Summary

Pending Requests

1

Active Rentals

0

Overdue

0

Rental Requests

Pending

Approved

Denied

All

REQ #	ASSET	CATEGORY	REQUESTED BY	START	DUE	ACTION
1	Iphone	Technology	John Smith (faculty)	2026-05-04	2026-05-05	Approve Deny

Active Rentals

RENTAL #	ASSET	RENTED BY	START	DUE	STATUS	ACTION
No active rentals found.						

Admins will also be able to gather an overall report of total assets and history of rental requests in the Summary tab. They may also download the summary as a separate and modular file using the Print/ Export button.

Campus Asset Tracker
IT Management Portal

admin1 Admin Manage Users Logout

Asset Inventory

Manage all campus assets and rental requests

Inventory Rental Management Summary

Asset Summary Report Generated 2026-05-03 18:34:55

Print / Export

Total Assets
2

Available
2

In Use
0

Rented
0

Maintenance
0

All Assets 2

ID	NAME	CATEGORY	SERIAL	LOCATION	STATUS
1	Laptop1	IT equipment	1234ABCDE	Computer Lab	Available
2	Iphone	Technology		Test	Available

To add or remove users, admins may use the “Manage Users” button at the top of the page.

Campus Asset Tracker
IT Management Portal

admin1 Admin Manage Users Logout

Asset Inventory

Manage all campus assets and rental requests

Inventory Rental Management Summary

Total Assets
2

Available
2

In Use
0

Maintenance
0

All Assets

All Categories All Statuses Sort

ID	NAME	CATEGORY	SERIAL NUMBER	STATUS	LOCATION
1	Laptop1	IT equipment	1234ABCDE	Available	Computer Lab
2	Iphone	Technology		Available	Test

Showing 2 of 2 assets

All active Student and Faculty users will be listed. On the left, admins will be able to create a new user. Deleting a user may be done with thin the actions column on the right.

Manage Users

+ Create User

Students

ID	Email	Name	Cell	Action
1	student1@gmail.com	Jane Doe	651-555-1234	Delete
2	student2@gmail.com	student student		Delete

Faculty

ID	Email	Name	Department	Action
1	teacher1@gmail.com	John Smith	Computer Science	Delete

Upon clicking the create user button, admins will be taken to the create user page. After filling out the appropriate information and clicking “Create User”, the user will now appear in the database and in the “Manage Users” page.

Create User

User Type:
Student

Email:
student3@gmail.com

Password:

First Name:
student3

Last Name:
student3

Cell:

Create User Cancel

Student/Faculty Logon

Students and Faculty will be met with a modified version of the dashboard. They will be limited to seeing only available items for rental, their rental requests, and active rentals.

Campus Asset Tracker
Equipment Portal

Jane Doe Student Logout

Equipment Rental
Browse and request available campus equipment

Available Equipment
2

My Active Rentals
0

Pending Requests
0

Available Equipment 2

My Requests

My Rentals

Available Equipment

Search by name, serial, or location...

All Categories

ID	NAME	CATEGORY	SERIAL	STATUS	LOCATION	ACTION
1	Laptop1	IT equipment	1234ABCDE	Available	Computer Lab	Request Rental
2	Iphone	Technology		Available	Test	Request Rental

2 available item(s)

Using the blue “Request Rental” button, students/faculty will be able to select a rental timeframe for the requested item.

Campus Asset Tracker
Equipment Portal

Jane Doe Student Logout

Equipment Rental
Browse and request available campus equipment

Available Equipment
2

My Active Rentals
0

Pending Requests
0

Available Equipment 2

My Requests

My Rentals

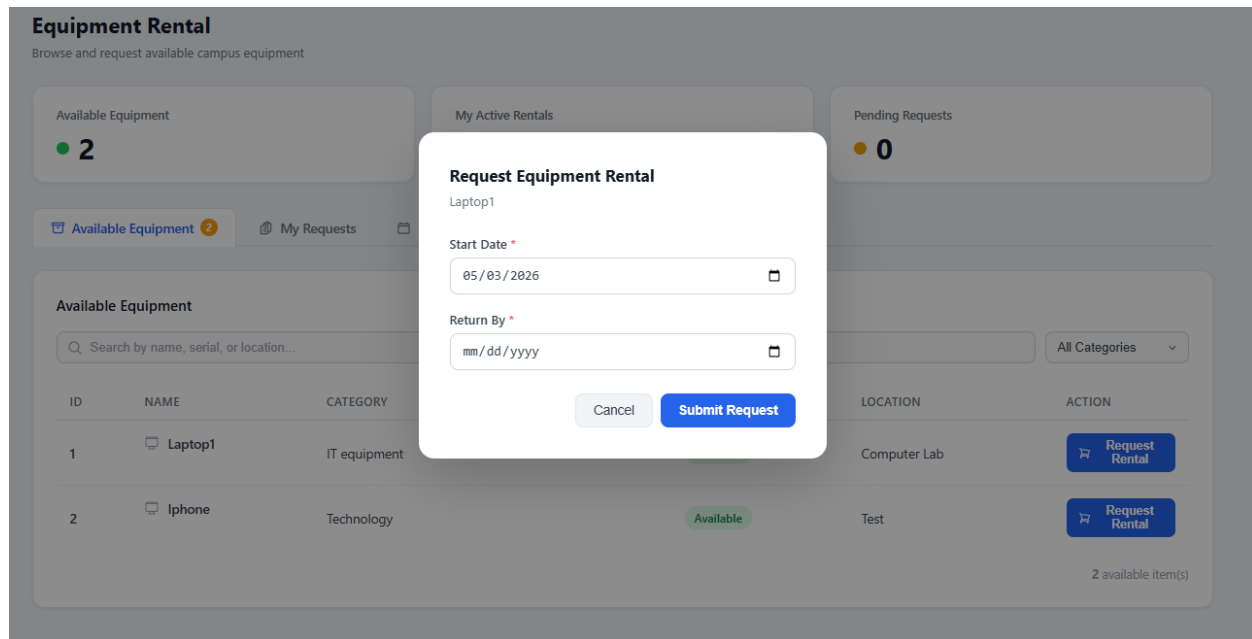
Available Equipment

Search by name, serial, or location...

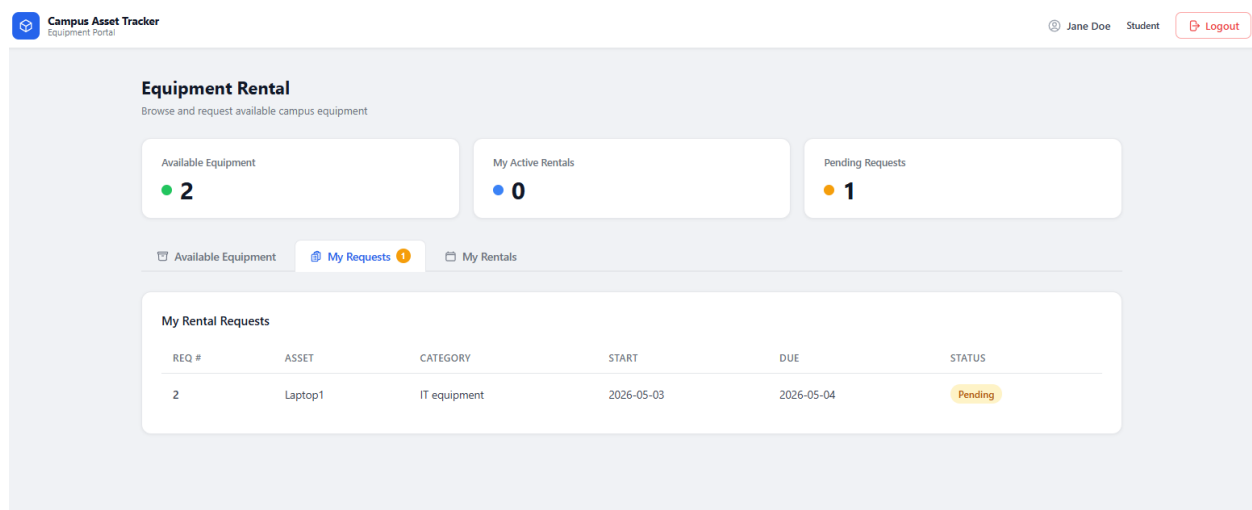
All Categories

ID	NAME	CATEGORY	SERIAL	STATUS	LOCATION	ACTION
1	Laptop1	IT equipment	1234ABCDE	Available	Computer Lab	Request Rental
2	Iphone	Technology		Available	Test	Request Rental

2 available item(s)



Once the request is submitted, the student/faculty will see a pending notification on the dashboard.



Once the rental request has been approved by an admin, the status of the rental request will update within the dashboard. A “Return Item” button will now be available to return and mark the item as “available” within the system.

Equipment Rental

Browse and request available campus equipment

Available Equipment

● 1

My Active Rentals

● 1

Pending Requests

● 0

📅 Available Equipment

📅 My Requests

📅 My Rentals 1

My Active Rentals

RENTAL #	ASSET	START	DUE	STATUS	ACTION
2	Laptop1	2026-05-03	2026-05-04	Active	Return Item

Index

Error: "MySQL shutdown unexpectedly. This may be due to a blocked port..."

- Resolved by going into the command line, use: `tasklist | findstr sql`, then `taskkill /F /PID PID_of_sql_tasks`

- Could also be a corrupted database. Database may require resetting and re-importing